Status	Finished
Started	Sunday, 18 October 2026, 5:36 PM
Completed	Sunday, 18 October 2026, 5:53 PM
Duration	17 mins 24 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **2**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 1000
c. 500
d. 100

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 99 ✓
d. 90 and 94

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By using a hash function
- b. By comparing each element in the list sequentially ✓
- c. By dividing the list into halves
- d. By sorting the list first

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search starts over from the beginning
- b. The search stops immediately ✓
- c. The search backtracks to find duplicate elements
- d. The search continues until the end of the list

Question **6**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Bubble ✓
- b. Complexity
- c. Selection
- d. Insertion

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search does not require dividing the list
- c. Binary search works on unsorted lists
- d. Binary search can find multiple instances of the target element

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Skipping every second element
- b. Sorting the list before searching
- c. Dividing the list in half repeatedly
- d. Checking each element sequentially ✓

Question **9**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Linear
- c. Binary ✓
- d. Both (1) & (3)

Question **10**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. In the middle of the list
- c. At the beginning of the list
- d. In the left sublist

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the left sublist ✓
c. In the right sublist
d. In the entire list

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. 1
d. n ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Hashing
c. Binary search
d. Linear search ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It works on unsorted lists

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers